

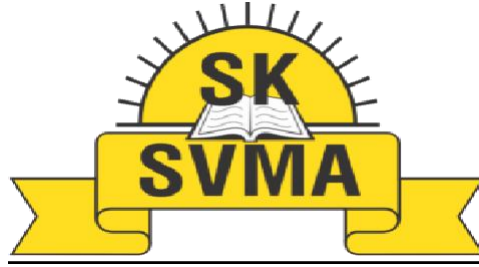


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ಶ್ರೀಮತಿ ಕಮಲಾ ಮತ್ತು ಶ್ರೀ ವೆಂಕಪ್ಪ ಎಂ. ಆಗಡಿ ಆಧಿಯಾಂತ್ರಿಕ ಮತ್ತು ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ, ಲಕ್ಷ್ಮೇಶ್ವರ
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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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SYNOPSIS
ON
MINI PROJECT
2025-26

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PROJECT TITLE

RFID Based Attendance System with Google Sheets using ESP8266

INTRODUCTION

Attendance tracking is an essential task in schools, colleges, and workplaces. Traditional methods, such as manual registers or roll calls, are time-consuming, prone to human errors, and often lack real-time data access. Automation using modern technologies provides an efficient and reliable solution.

This project, **RFID-Based Attendance System with Google Sheets using ESP8266**, is designed to automate attendance management using **Radio Frequency Identification (RFID) technology**. Each student or employee is assigned an RFID card with a unique identification number. When the card is scanned by the RFID reader, the data is transmitted to the **ESP8266 Wi-Fi module**, which connects to the internet and updates the attendance record directly in **Google Sheets** in real time.

By integrating Google Sheets, the system ensures that attendance data is **securely stored, easily accessible, and shareable** with authorized personnel. Teachers, administrators, or managers can view and analyze records instantly, eliminating manual entry.

This system not only **saves time and reduces paperwork** but also **improves accuracy and transparency**. It can be deployed in schools, colleges, offices, libraries, and other organizations requiring **efficient attendance management**.

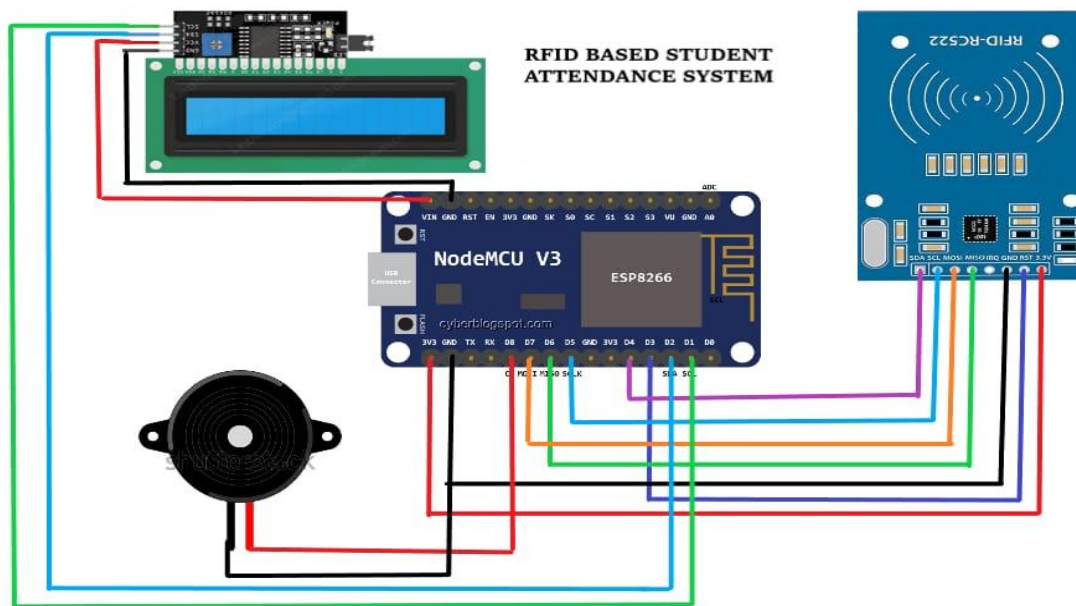
LITERATURE SURVEY

- **RFID-based attendance systems** — Widely used for contactless, fast identification; commonly implemented with MFRC522 or similar readers and passive RFID cards for low-cost roll-call automation.
- **Microcontroller integration (ESP8266/Arduino)** — ESP8266 is popular for Wi-Fi enabled IoT projects due to low cost and easy MQTT/HTTP support; many studies show it reliably forwards sensor/tag reads to cloud endpoints.
- **Cloud logging using Google Sheets / Apps Script** — Google Sheets offers an accessible cloud backend via Google Apps Script or REST APIs; prior works use it for real-time logging and simple dashboards without needing dedicated servers.
- **Security & privacy concerns** — Literature highlights vulnerabilities (cloned tags, unencrypted transmission); recommended mitigations include tag UID validation, HTTPS, and simple server-side checks to prevent spoofing.
- **Comparison with biometrics (fingerprint/face)** — RFID is faster and cheaper but less secure than biometric methods; trade-offs are cost, user convenience, and spoof-resistance.
- **System constraints reported** — Common limitations: offline behavior when Wi-Fi fails, duplicate scans, limited scalability for very large deployments, and lack of robust access control in basic implementations.
- **Gap addressed by this project** — Combines low-cost RFID + ESP8266 with Google Sheets for immediate, portable cloud logging; focuses on simplicity, real-time access, and an easy-to-deploy solution suitable for small-to-medium institutions while noting security improvements (UID checks, basic HTTPS/API use).

OBJECTIVES

- **Automate Attendance** – Replace manual roll calls with an RFID-based system for fast and accurate attendance marking.
- **Enable Real-Time Logging** – Use ESP8266 Wi-Fi to send attendance data instantly to Google Sheets.
- **Provide Cloud Storage** – Maintain attendance records securely on Google Sheets for easy access and analysis.
- **Enhance Transparency** – Ensure accurate, tamper-proof records accessible to teachers, administrators, or managers.
- **Reduce Administrative Workload** – Minimize manual errors and save time in maintaining attendance registers.
- **Ensure Scalability & Low Cost** – Develop a cost-effective solution suitable for schools, colleges, offices, and events.
- **Support Remote Monitoring** – Allow administrators to view attendance data anytime, anywhere via cloud access.

METHODOLOGY



COMPONENTS REQUIRED

HARDWARE

- ESP8266 NodeMCU
- RFID Reader (MFRC522 Module)
- RFID Tags/Cards
- Jumper Wires
- Breadboard
- Power Supply / USB Cable
- LED/Buzzer (optional)

SOFTWARE

- Arduino IDE
- ESP8266 Board Package & Libraries .
- Google Sheets Google Apps Script
- Web Browser

EXPECTED OUTCOME

- **Automated Attendance System** – Attendance of students/employees is marked instantly by scanning RFID cards.
- **Real-Time Data Logging** – Attendance records are uploaded directly to Google Sheets through ESP8266.
- **Cloud-Based Storage** – Secure, easily accessible records stored online without need for local databases.
- **Error-Free Records** – Eliminates manual entry errors and reduces chances of proxy attendance.
- **User-Friendly Interface** – Simple access to attendance sheets via Google Sheets for teachers/admins.
- **Low-Cost & Scalable Solution** – Affordable hardware and open-source software suitable for schools, colleges, and offices.
- **Remote Monitoring** – Attendance data can be viewed anytime, anywhere using internet-connected devices.
